NOURDDINE BOUAZIZ

PhD in Computer Science (Arts et Métiers) – Postdoctoral Researcher, CESI LINEACT
Operations Research & Scheduling • Simulation • Python & Odoo
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PROFILE

PhD in Computer Science (Arts et Métiers, 2025) with an Industrial Engineering background. Qualified for French associate professor positions (CNU sections 27 and 61, 2026). I build decision-support tools for manufacturing: **modelling, simulation and optimisation**. Postdoctoral researcher at CESI LINEACT working on the scheduling of a robotic machining cell (OPTIMAN project, with Maugars Industrie). My career combines research on human-behaviour uncertainty in scheduling (five publications, including IJPE 2024), **Python** development – main author of the open-source **Netlogopy** library – and five years of **Odoo** ERP development in port logistics, manufacturing and point of sale (2017–2022).

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher – Scheduling of a robotic machining cell *09/2025 – present*
CESI LINEACT • OPTIMAN project (with Maugars Industrie; Normandy Region / ERDF) *Rouen, France*

- Formalised the cell constraints with the industrial partner: 2 CNC machines, robot, pallets, tooling, human interventions and operator calendars.
- **Python planning tool**: validated Excel import, constructive heuristic under resource constraints, JSON API, machine / robot / operator Gantt charts, **Odoo 12** connector.
- Scheduling optimisation with a **genetic algorithm** and integration with Odoo – manuscript in preparation.

PhD Researcher – Scheduling and human behaviour *04/2022 – 04/2025*
CESI LINEACT / Arts et Métiers • AntiHpert project (RIN Tremplin 2019) *Rouen, France*

- Thesis defended on 25 June 2025 (supervisors M. Sahnoun and A. Yassine): *Predictive dynamic scheduling of a flexible production workshop under human-behaviour uncertainty.*
- **Stochastic modelling** of operators (Markov chains, fatigue, learning) and **agent-based simulation** in NetLogo/Python of a dual-resource-constrained flow shop.
- **MINLP optimisation** (Gurobi and SCIP under Julia) of worker-to-workstation assignment; simulation–optimisation loop triggering rescheduling on deviation from the plan.
- **Netlogopy**: open-source Python–NetLogo library (MIT, PyPI). **FrogsNBots**: shop-floor data collection (Odoo, QR scans, synchronised multi-camera video).
- Systematic review of 41 papers; teaching (95 h tutorial/lab equivalent) and co-supervision of student projects.

ERP (Odoo) and Application Developer *02/2020 – 03/2022*
SMARTSTUP Group *Algeria*

- **Odoo** manufacturing (MRP), point-of-sale and restaurant modules; IoT peripheral integration (Raspberry Pi); web, mobile and desktop applications.
- Simulation and optimisation of industrial flows; user training and documentation.

Freelance ERP Consultant and Developer *08/2018 – 02/2020*
Manufacturing and food-industry clients *Remote*

- Tempered-glass factory: **Odoo** pricing and **cutting-stock** modules, linked to order processing.
- Food production–distribution network: multi-stage non-linear model (LINGO, CPLEX).

Odoo Developer *01/2017 – 08/2018*
Port Authority of Ghazaouet *Ghazaouet, Algeria*

- Process analysis (**MERISE/UML**) and development of four **Odoo** modules: CMMS, invoicing, operations, sales; user training.
- **Berth allocation** (OpenSolver, LINGO, Arena simulation; solution implemented by the port) and container storage (genetic algorithms).

EDUCATION

PhD in Computer Science *2022 – 2025*
École Nationale Supérieure d'Arts et Métiers (HESAM Université), Paris – defended 25 June 2025, unanimous jury

Qualification for associate professor positions (maître de conférences) *2026*
French National Council of Universities (CNU), sections 27 (Computer Science) and 61 (Computer Engineering, Automatic Control and Signal Processing) – valid until 31/12/2030

Master's in Industrial Engineering, Production Engineering track *2015 – 2017*
University of Tlemcen, Algeria – project: CMMS and berth allocation (Port of Ghazaouet)

Bachelor's in Industrial Engineering *2012 – 2015*
University of Tlemcen, Algeria – project: food supply-chain planning (LINGO, genetic algorithm in Java)

PUBLICATIONS

[1] Bouaziz, N., Bettayeb, B., Sahnoun, M., Yassine, A. (2024). Incorporating uncertain human behavior in production scheduling for enhanced productivity in Industry 5.0 context. *International Journal of Production Economics*, 274, 109311. doi:10.1016/j.ijpe.2024.109311

[2] Bouaziz, N., Bettayeb, B., Sahnoun, M., Yassine, A. (2025). Netlogopy: Unlocking Advanced Simulation and Integration for NetLogo Using Python. *Simulation Workshop 2025 (SW25)*, Exeter, UK. doi:10.36819/SW25.019

[3] Bouaziz, N., Bettayeb, B., Sahnoun, M., Yassine, A. (2024). Modeling and simulation of worker profiles in production systems based on punctuality, fatigue, and experience. *7th IEEE Int. Conf. on Technology Management, Operations and Decisions (ICTMOD 2024)*, Sharjah, UAE. doi:10.1109/ICTMOD63116.2024.10878184

[4] Bouaziz, N., Bettayeb, B., Sahnoun, M., Yassine, A., Latreche, A. (2022). Modeling and simulation of human behavior impact on production throughput. *10th IFAC MIM 2022, IFAC-PapersOnLine*, 55(10), 1740–1745. doi:10.1016/j.ifacol.2022.09.649

[5] Bouaziz, N., Bettayeb, B., Sahnoun, M., Yassine, A. (2024). Ordonnancement d'atelier de production flexible sous incertitude du comportement humain. *2nd annual SAGIP congress*, Villeurbanne, France, May 2024. hal-04627782

Three journal articles submitted in 2025 (systematic review, Netlogopy, dynamic simulation–optimisation).

SOFTWARE AND OPTIMISATION PROJECTS (SELECTION)

- **Netlogopy** – open-source Python–NetLogo library (MIT, PyPI, GitHub): direct creation and control of agents at runtime, reuse of existing models, integration with the Python ecosystem.
- **FrogsNBots** – multimodal dataset of industrial actions: Odoo module, QR scans, synchronised fixed and PTZ cameras, JSON annotations.
- **Flexibac** (2026) – MILP model and simulation–optimisation loop with FlexSim for robotic parcel sorting (assignment to carts, capacities, time windows).
- **Logistics optimisation** – butane-distribution VRP/CVRP (Naftal, Tlemcen: LINGO + genetic algorithm in Java); multi-period aggregate planning; workshop scheduling (single machine, parallel machines, flow shop); facility location (p-median, fuzzy logic).

SKILLS

- **Operations research:** MILP / MINLP modelling, scheduling (flow shop, flexible job shop, dual resource constraints), assignment, VRP, planning; metaheuristics (genetic algorithms, NSGA-II); solvers Gurobi, SCIP, CPLEX, LINGO, OpenSolver, OR-Tools.
- **Simulation:** agent-based (NetLogo, Netlogopy, Python), discrete-event (Arena, FlexSim), Markov chains, simulation–optimisation.
- **Development:** Python (Flask, Django, pandas, Pyomo, Pymoo), Julia, Java, SQL / PostgreSQL, JavaScript; Odoo v9 to v16 (ORM, QWeb views, API, business modules); Git, Docker, Linux.
- **Data and domain:** MERISE/UML, production data (Excel, CSV, JSON, APIs), CMMS, MRP, POS, port logistics, Industry 4.0 / 5.0.
- **Languages:** French and English (professional fluency), Arabic (native).